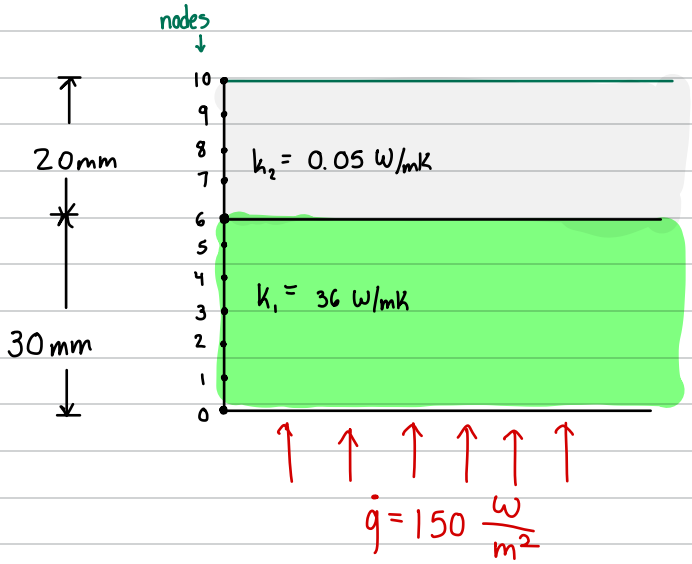
Problem 1

Diagram:

Air

$T_\infty = 15°C, h = 5 W/m^2K$



$\Delta X = 5mm$

Assumptions:

- Steady State
- One dimensional - x
- Uniform properties:
  h = const, k = const, $\dot{q}$ = const, temperatures
- Radiation is negligible
- Mecanismos: Conducción y convección
- No heat generation

Procedure:

$\frac{L}{\Delta X} + 1 = M$

$\frac{(20mm + 30mm)}{5mm} + 1 = 11$ nodes

Boundary Conditions

m=10: $\dot{Q} = A_s h (T - T_\infty)$

$\dot{q} = (T_{10} - T_\infty) h$

$\dot{q}/h + T_\infty = T_{10}$

$150 \frac{W}{m^2} / 5 \frac{W}{m^2 K} \left(\frac{K}{°C}\right) + 15°C = T_{10}$

$30°C + 15°C = 45°C$

$\dot{q} = \frac{\dot{Q}}{A}$ $\quad A\dot{q} = \dot{Q}$

$\dot{q}_0 + k_1 \frac{(T_1 - T_0)}{\Delta X} = 0$ $\quad \Delta T_{k=36 \frac{W}{mK}}$

$150 \frac{W}{m^2} + 36 \frac{W}{mK} \left(\frac{1K}{°C}\right) \cdot \frac{(\Delta T)}{0.005m} = 0$

$\Delta T_{k=36 \frac{W}{mK}} = -0.02083°C$

$150 \frac{W}{m^2} + 0.05 \frac{W}{mK} \left(\frac{1K}{°C}\right) \cdot \frac{(\Delta T)}{0.005m} = 0$

$\Delta T_{k=0.05 \frac{W}{mK}} = -15°C$

$\dot{Q} = -\frac{kA(T_1 - T_0)}{\Delta X}$

$\dot{q}_0 = -\frac{k_1 (T_1 - T_0)}{\Delta X}$

$m = 6$: $150 \frac{W}{m^2} = -36 \frac{W}{mK}\left(\frac{1K}{°C}\right) \cdot \frac{\Delta T_1}{0.03m}$

$\Delta T_1 = -0.125\,°C$

$m = 10$: $150 \frac{W}{m^2} = -0.05 \frac{W}{mK}\left(\frac{1K}{°C}\right) \cdot \frac{\Delta T_2}{0.02m}$

$\Delta T_2 = -60\,°C$

$m = 0$: $\Delta T = \Delta T_1 + \Delta T_2$

$T_{10} - T_0 = \Delta T$

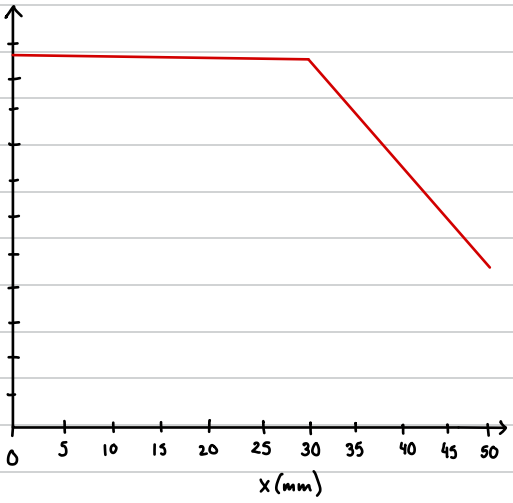
$T_{10} - (\Delta T_1 + \Delta T_2) = T_0$

$45°C - (-0.125°C - 60°C) = 105.125°C$

$T_0 = 105.125\,°C$

Since $T_0 = 105.125\,°C$ and $T_0 > 93°C$, the ASME SB-96 plate is not compliant with the ASME Boiler and Pressure Vessel Code, thus it cannot be operated.

Tempevature Profile:



m	X (mm)	T_m (°C)	
0	0	105.125	
1	5	105.10417	
2	10	105.08334	$\Delta T_{k=36} = -0.02083°C$
3	15	105.06251	
4	20	105.04168	
5	25	105.02085	
6	30	105.00002	
7	35	90	
8	40	75	$\Delta T_{k=0.05} = -15°C$
9	45	60	
10	50	45	

Complete procedure instead of simply applying ΔT_{ki} from each material. Same results are obtained. For example:

$m = 1$: $T_0 - 2T_1 + T_2 = 0$

$T_1 - T_0 = -0.02083°C$

$T_1 = -0.02083 + T_0$

$m = 2$ $T_0 - 2(-0.02083°C + T_0) + T_2 = 0$

$T_2 = 2(-0.02083 + 105.125°C) - 105.125°C$

$T_2 = 105.083°C$

$\frac{T_2 + T_0}{2} = T_1 \quad \therefore \quad T_1 = \frac{105.083°C + 105.125°C}{2} = 105.104\,°C$

Finite Difference method

Σ heat into node per unit area $= 0$

$$\frac{k_1}{\Delta x} = \frac{36\ W/mK}{0.005m} = 7200\ \frac{W}{m^2K} \qquad \frac{k_2}{\Delta x} = \frac{0.05\ W/mK}{0.005m} = 10\ W/m^2K$$

$m = 0$

$$150\ \frac{W}{m^2} + 7200\ \frac{W}{m^2K}(T_1 - T_0) = 0$$

interior nodes

$$\frac{k(T_{m-1} - T_m)}{\Delta x} + \frac{k(T_{m+1} - T_m)}{\Delta x} = 0 \qquad \text{Divide by } \frac{k}{\Delta x}$$

$$T_{m-1} - 2T_m + T_{m+1} = 0 \qquad \text{for } m = 1\text{-}5,\ m = 7\text{-}9$$

$m = 6$ (interface)

$$7200\ \frac{W}{m^2K}(T_5 - T_6) + 10\ \frac{W}{m^2K}(T_7 - T_6) = 0$$

$m = 10$

$$10\ \frac{W}{m^2K}(T_9 - T_{10}) + 5\ \frac{W}{m^2{}^\circ C}(15\,^\circ C - T_{10}) = 0$$

Solve system

$$\dot{q} = (T_{10} - T_\infty)h \qquad 150\ \frac{W}{m^2} \Big/ 5\ \frac{W}{m^2K}\left(\frac{K}{{}^\circ C}\right) + 15\,^\circ C = T_{10}$$

$$\dot{q}/h + T_\infty = T_{10} \qquad 30\,^\circ C + 15\,^\circ C = 45\,^\circ C = T_{10}$$

Same as first method.

$$T_9 \rightarrow 10\frac{W}{m^2{}^\circ C}(T_9 - 45\,^\circ C) + 5\frac{W}{m^2{}^\circ C}(15\,^\circ C - 45\,^\circ C) = T_9 = 60\,^\circ C$$

$T_8 \rightarrow T_8 = 2T_9 - T_{10} \Rightarrow 2(60\,^\circ C) - 45\,^\circ C = 75\,^\circ C = T_8$

$T_7 \rightarrow T_7 = 2(75\,^\circ C) - 60\,^\circ C = 90\,^\circ C = T_7$

$T_6 \rightarrow T_6 = 2(90\,^\circ C) - 75\,^\circ C = 105\,^\circ C = T_6$

$T_5 \rightarrow 7200\ \frac{W}{m^2{}^\circ C}(T_5 - 105\,^\circ C) + 10\frac{W}{m^2{}^\circ C}(90\,^\circ C - 105\,^\circ C) = 0 \therefore$

$T_5 = 105.02083\,^\circ C$

$T_4 \rightarrow T_4 = 2(105.0283\,^\circ C) - 105\,^\circ C = 105.042\,^\circ C = T_4$

$T_3 \rightarrow T_3 = 2(105.042\,^\circ C) - 105.0208\,^\circ C = 105.062\,^\circ C = T_3$

$T_2 \rightarrow T_2 = 2(105.0625\,^\circ C) - 105.042\,^\circ C = 105.083\,^\circ C = T_2$

$T_1 \rightarrow T_1 = 2(105.0833\,^\circ C) - 105.0625\,^\circ C = 105.104\,^\circ C = T_1$

$T_0 \rightarrow T_0 = 2(105.1042\,^\circ C) - 105.0833\,^\circ C = 105.125\,^\circ C = T_0$

m	X (mm)	T_m (°C)
0	0	105.125
1	5	105.104
2	10	105.083
3	15	105.062
4	20	105.042
5	25	105.021
6	30	105.000
7	35	90
8	40	75
9	45	60
10	50	45

Lowest h value so allow does not exceed 93°C.

$$\Delta T = \Delta T_1 + \Delta T_2$$
$$\Delta T = -0.125°C - 60°C$$
$$\Delta T = -60.125°C$$

$$T_{10} - (\Delta T_1 + \Delta T_2) = T_0$$
$$T_{10} - (-60.125°C) < 93°C$$
$$T_{10} < 32.875°C$$

$$\dot{q} = (T_{10} - T_\infty) h$$
$$\dot{q}/h + T_\infty = T_{10}$$
$$\frac{\dot{q}}{h} + T_\infty < 32.875°C$$
$$\frac{\dot{q}}{h} < 32.875°C - T_\infty$$
$$150 \frac{W}{m^2} / (32.875°C - 15°C) < h$$
$$8.39161 \frac{W}{m^2 °C} < h$$

$$h > 8.39161 \frac{W}{m^2 K}$$